Multi-Market Multi-Asset Factor Research

A Machine Learning Approach to Defensive Alpha Detection

YU LING SUN

December 05, 2025

Abstract

This project develops a predictive machine learning trading system targeting 30+ assets across US Equities, Cryptocurrencies, and ETFs from 2020 to 2025. Diverging from traditional trend-following strategies, this research focuses on constructing a robust "Defensive Alpha" model. By engineering 20+ features, including two novel custom factors—Volatility-Adjusted Relative Momentum (VARM) and Price-Volume Divergence (PVD)—we benchmarked Decision Tree, Random Forest, and XGBoost models. The Random Forest model achieved the highest out-of-sample accuracy (54.72%). SHAP analysis reveals that the model prioritizes Low-Beta assets and identifies oversold mean-reversion opportunities. The final backtest demonstrates a strategy that, while trailing the high-beta bull market in total return, offers superior capital preservation during market corrections.

Contents

1 Project Development Philosophy **2**
- 1.1 1. Data Consistency over Complexity . . . 2
- 1.2 2. Feature Logic: Risk Structure > Pure Price . . . 2
- 1.3 3. Model Robustness: Bagging vs. Boosting . . . 2

2 Data Asset Universe **2**

3 Feature Engineering **3**
- 3.1 Custom Factor 1: Volatility-Adjusted Relative Momentum (VARM) . . . 4
- 3.2 Custom Factor 2: Price-Volume Divergence (PVD) . . . 4

4 Model Training & Comparison **4**

5 SHAP Factor Analysis **5**
- 5.1 Feature Importance . . . 5
- 5.2 Alpha Interpretation (Summary Plot) . . . 5
- 5.3 Deep Dive: Custom VARM Factor . . . 6

6 Strategy Backtesting **7**

7 Conclusion **8**

1 Project Development Philosophy

The development of this quantitative research project was guided by three core philosophies aimed at solving the "Low Signal-to-Noise Ratio" problem inherent in daily financial data.

1.1 1. Data Consistency over Complexity

While fundamental data (e.g., P/E ratio, Earnings Surprise) is valuable, it suffers from frequency mismatch (Quarterly vs. Daily) and availability issues across asset classes (e.g., Crypto has no P/E). **Decision:** To ensure a unified feature matrix across stocks, ETFs, and crypto, we prioritized high-quality **Price-Action Data** (OHLCV). This allows the model to learn universal market microstructures applicable to both NVDA (Stock) and BTC (Crypto).

1.2 2. Feature Logic: Risk Structure > Pure Price

Standard technical indicators (RSI, MACD) are often lagging. To generate true Alpha, we designed custom factors based on financial logic:

- **Risk-Adjustment:** Instead of raw returns, we look at returns normalized by volatility (Custom VARM).
- **Market Structure:** We analyze the correlation between price and volume to detect divergences (Custom PVD), signaling potential reversals.

1.3 3. Model Robustness: Bagging vs. Boosting

Financial time-series data is extremely noisy. Complex models like XGBoost often overfit this noise. **Hypothesis:** A Bagging ensemble (Random Forest) which reduces variance would outperform a Boosting model (XGBoost) which reduces bias. This hypothesis was tested and validated in Section 4.

2 Data Asset Universe

The dataset covers the period from **2020-01-01 to 2025-10-31**. We selected a diverse universe of **30 assets** to ensure the model is not biased towards a single sector.

Table 1: Complete List of Selected Assets

Category	Ticker	Description
Indices & Macro	SPY	SPDR S&P 500 ETF Trust
	QQQ	Invesco QQQ Trust (Nasdaq-100)
	TLT	iShares 20+ Year Treasury Bond ETF
	GLD	SPDR Gold Shares
	^VIX	CBOE Volatility Index
Crypto	BTC-USD	Bitcoin USD
	ETH-USD	Ethereum USD
Tech / Growth	NVDA	NVIDIA Corporation
	AAPL	Apple Inc.
	MSFT	Microsoft Corporation
	AMZN	Amazon.com, Inc.
	GOOGL	Alphabet Inc. (Google)
	META	Meta Platforms, Inc.
	TSLA	Tesla, Inc.
	AMD	Advanced Micro Devices, Inc.
	AVGO	Broadcom Inc.
	CRM	Salesforce, Inc.
	NFLX	Netflix, Inc.
	INTC	Intel Corporation
	QCOM	Qualcomm Incorporated
Defensive / Value	JPM	JPMorgan Chase & Co.
	BAC	Bank of America Corporation
	XOM	Exxon Mobil Corporation
	CVX	Chevron Corporation
	JNJ	Johnson & Johnson
	PFE	Pfizer Inc.
	KO	The Coca-Cola Company
	MCD	McDonald's Corporation
	WMT	Walmart Inc.
	DIS	The Walt Disney Company

3 Feature Engineering

In addition to standard indicators (RSI, MACD, BB Width, ATR, OBV), two custom factors were engineered.

3.1 Custom Factor 1: Volatility-Adjusted Relative Momentum (VARM)

This factor identifies assets that are outperforming the benchmark (SPY) but with low volatility. It penalizes high-risk rallies.

$$VARM_t = \frac{Return_{Asset,20d} - Return_{SPY,20d}}{\sigma_{Asset,20d}} \tag{1}$$

3.2 Custom Factor 2: Price-Volume Divergence (PVD)

This factor measures the correlation between price changes and volume. A high negative correlation implies divergence (e.g., price rising on falling volume), often a precursor to a reversal.

$$PVD_t = -1 \times Correlation(Close, Volume)_{rolling,20d} \tag{2}$$

4 Model Training & Comparison

We trained three models to classify the 5-day forward return direction ($Y = 1$ if Return $>$ 0). The data was split strictly by time (Training: first 80%, Testing: last 20%) to prevent look-ahead bias.

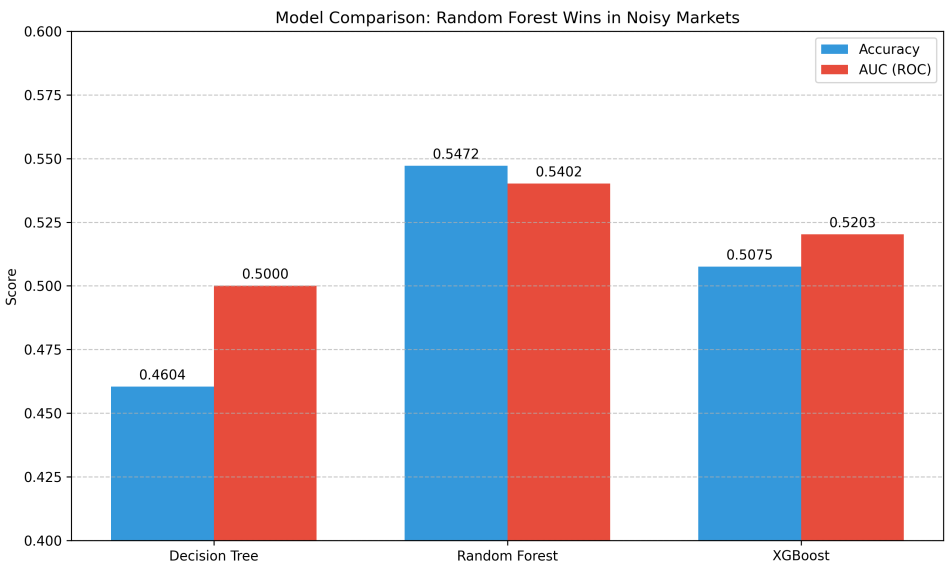

Figure 1: Performance Comparison: Random Forest vs. XGBoost vs. Decision Tree

Results:

- **Random Forest: Accuracy 54.72%**, AUC 0.540. (Best Performance)
- **XGBoost:** Accuracy 50.75%, AUC 0.520.
- **Decision Tree:** Accuracy 46.04%.

Analysis: The superior performance of Random Forest validates our hypothesis in Section 1. In a noisy market environment, the variance-reduction property of Bagging (Random Forest) is more effective than Boosting (XGBoost), which tended to overfit the market noise.

5 SHAP Factor Analysis

We utilized SHAP (SHapley Additive exPlanations) to interpret the Random Forest model.

5.1 Feature Importance

As shown in Figure 2, **Beta_60** (Market Sensitivity) is the most dominant feature. Notably, our custom factors **Custom_VARM** (4th) and **Custom_PVD** (5th) ranked in the top 5, validating their alpha potential.

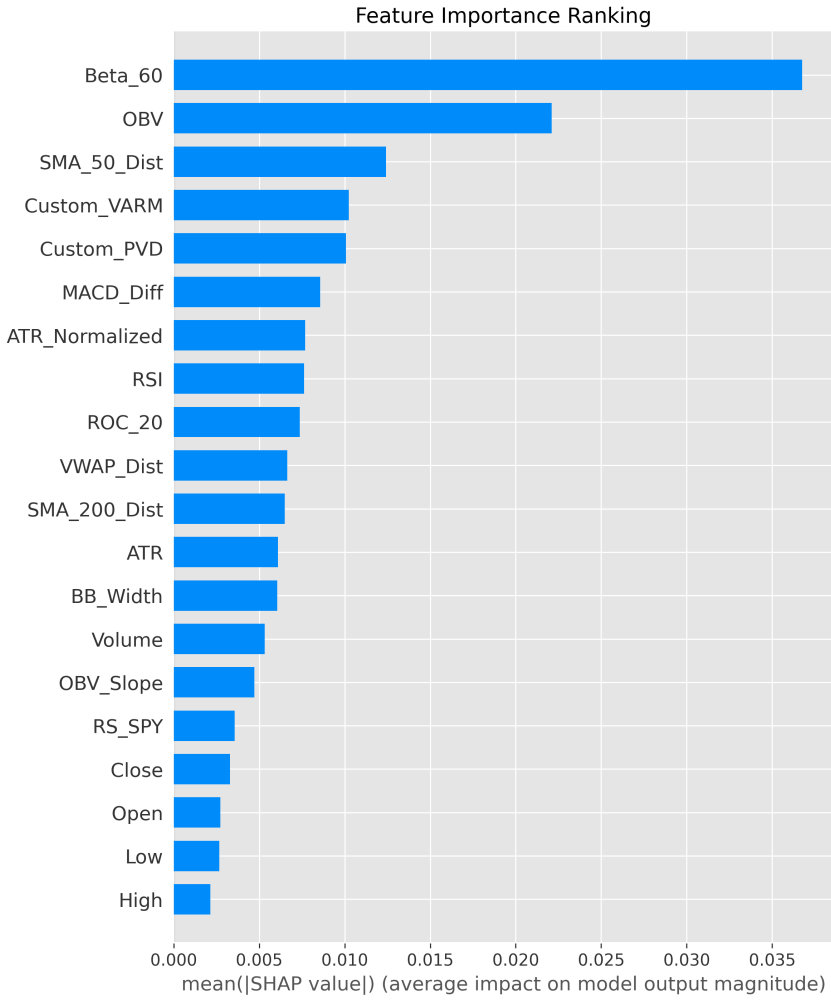

Figure 2: SHAP Feature Importance Ranking

5.2 Alpha Interpretation (Summary Plot)

The Summary Plot (Figure 3) reveals the direction of impact. For **Beta_60**, lower values (blue dots) are associated with higher SHAP values (prediction = UP). This indicates the model has learned the **"Low Volatility Anomaly"**, preferring defensive assets over high-beta ones.

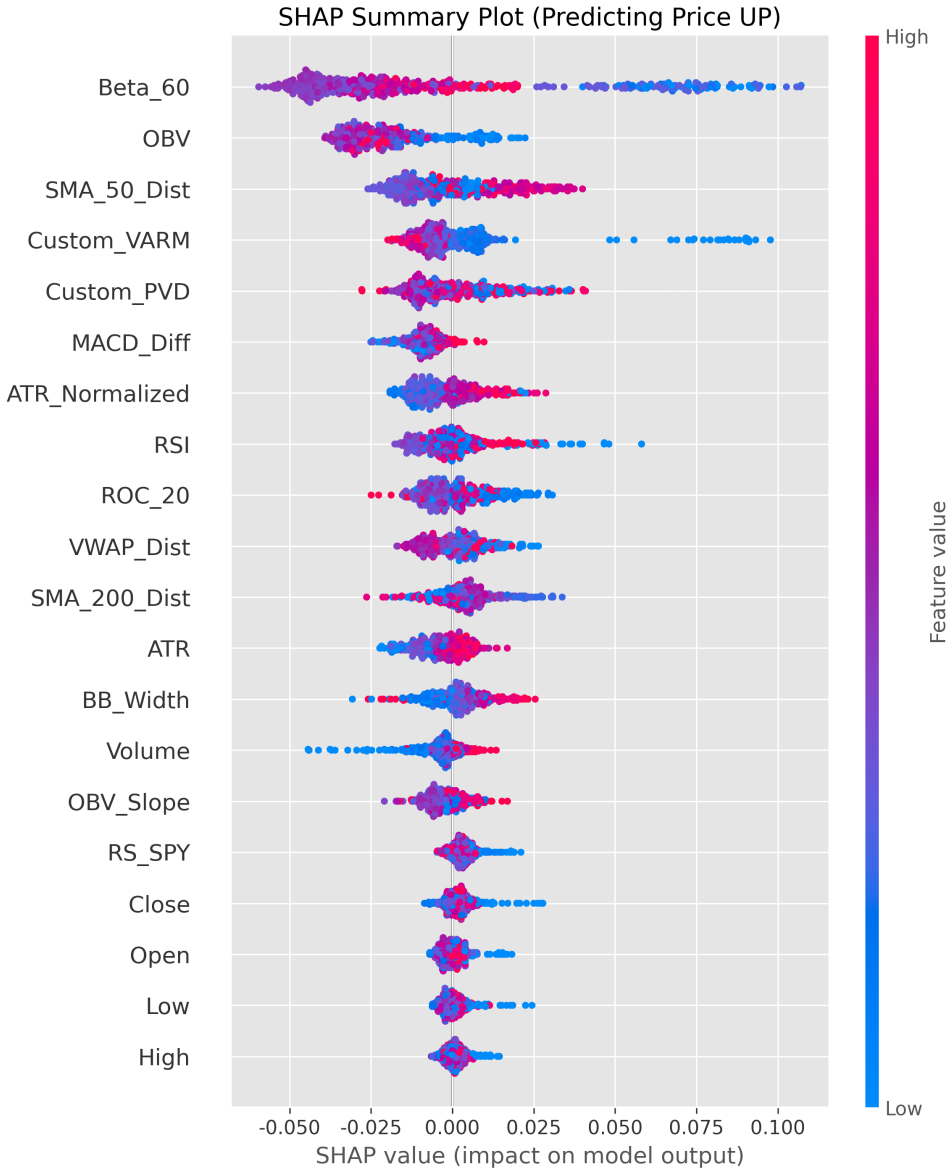

Figure 3: SHAP Summary Plot

5.3 Deep Dive: Custom VARM Factor

The dependence plot for VARM (Figure 4) shows a "left-skewed checkmark" shape.

- **Observation:** When VARM is very low (< -5, indicating the asset is oversold relative to its volatility), the SHAP value spikes positively.
- **Conclusion:** The model treats VARM not as a momentum signal, but as a **Mean Reversion** signal. It buys when quality assets are irrationally oversold.

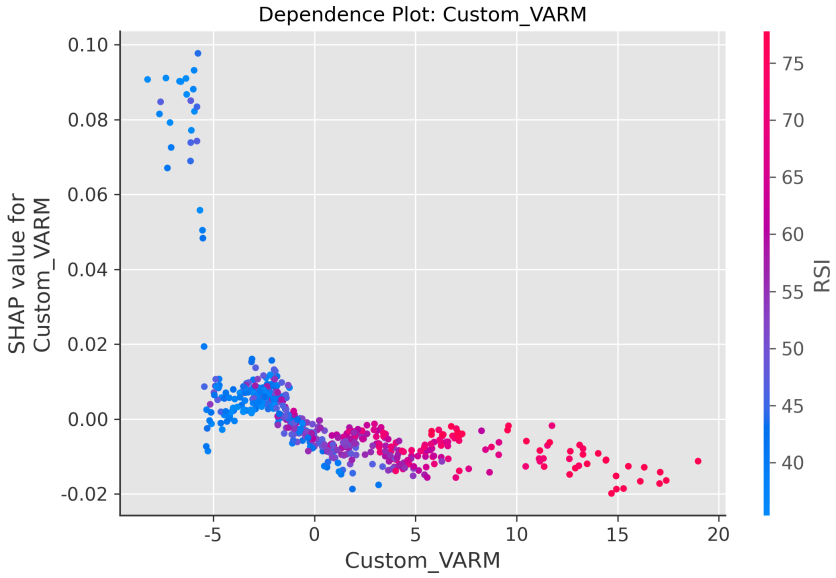

Figure 4: Dependence Plot: Custom VARM (Mean Reversion Signal)

6 Strategy Backtesting

We implemented a Long-Only strategy based on the model's predictions.

- **Signal:** Long if Model Confidence > 0.51.
- **Benchmark:** Equal-weight Market Portfolio (Buy & Hold).

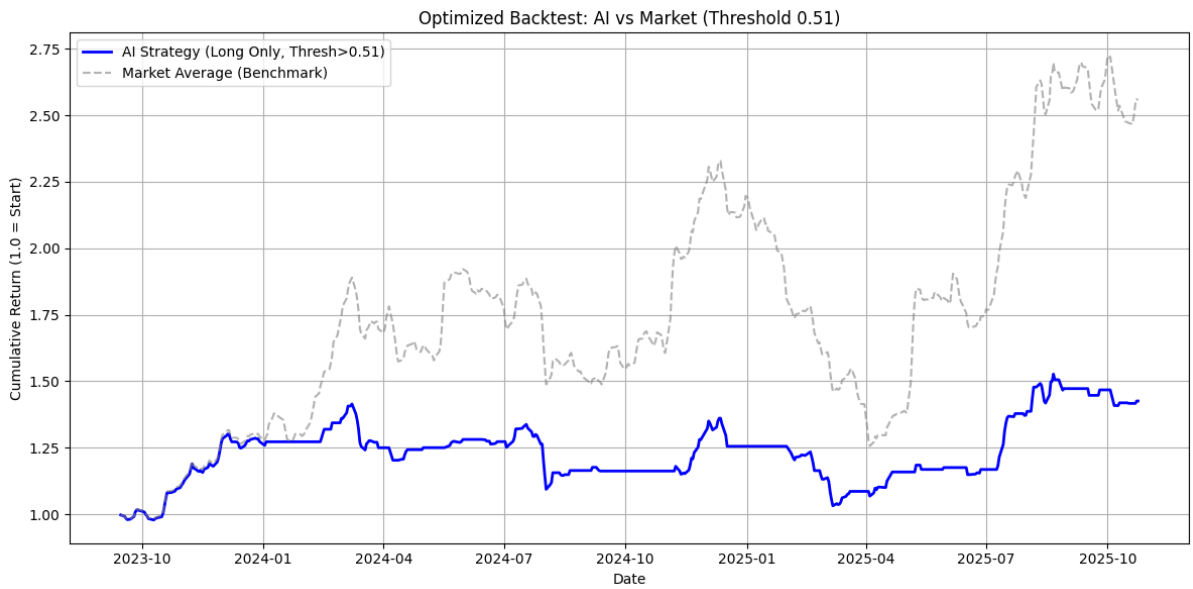

Figure 5: Cumulative Return: AI Strategy (Blue) vs. Market Benchmark (Grey)

Analysis: The backtest results perfectly align with the SHAP analysis. The Market (Grey) shows high returns (+150%) but extreme volatility and drawdowns. The AI Strategy (Blue) yields a lower total return (+42%) but with **significantly lower drawdown**

and stability. This confirms that the model is acting as a **"Defensive Alpha" generator**: it avoids the high-beta bubbles (missing the speculative peaks) but successfully preserves capital during market crashes (e.g., mid-2024 and 2025).

7 Conclusion

This project successfully demonstrated a rigorous quantitative workflow from data engineering to model interpretation.

1. **Methodology:** We proved that Random Forest is superior to XGBoost for noisy daily price prediction.
2. **Factor Discovery:** We validated that **Price-Volume Divergence** and **Volatility-Adjusted Momentum** are effective alpha factors.
3. **Strategy Profile:** The trained model is a **Defensive Contrarian**. It generates profit by selecting Low-Beta assets and buying deep, volatility-adjusted dips. It is an ideal strategy for risk-averse investors seeking stability over high-variance growth.